

Vikas Pratap Singh

+91 70148 57644 | vikas.iiitp@gmail.com | [LinkedIn](#)

Data Engineer

Python | PySpark | Apache Airflow | Hive | Hadoop | SQL | ETL | Distributed Data Processing

Data Engineer with 5 years of experience designing and optimizing large-scale distributed data systems using **PySpark, Hive, Apache Airflow, and Hadoop**. Proven track record of building high-performance ETL pipelines processing **100M+ records per week**, improving execution efficiency, reliability, and observability. Strong experience in distributed data transformation, performance tuning, pipeline architecture, and production system ownership in enterprise environments.

Associate Technical Architect (Data Engineering)

o9Solutions – Bengaluru

Apr 2025 – Present

- Architected scalable distributed ETL frameworks using **PySpark, Hive, and Apache Airflow**, supporting 5+ enterprise production implementations.
- Designed reusable Spark-based transformation patterns leveraging **DataFrame API, lazy evaluation, and optimized execution strategies** for large-scale datasets.
- Improved Spark job performance through **column pruning, broadcast joins, and repartitioning strategies to handle data skew**, reducing shuffle overhead in production pipelines.
- Designed centralized **data lineage and pipeline observability framework**, improving troubleshooting efficiency and system transparency.
- Provided technical leadership on distributed data architecture, mentoring engineers on Spark optimization and best practices.
- Leveraged AI-assisted documentation (NotebookLM) to analyze pipeline architectures, identify design gaps, and standardize technical documentation (TDDs, runbooks, governance frameworks).

Onsite Client Engagement – United States

Mar 2025

- Resolve a **critical production escalation in an enterprise planning platform** impacting business operations.
- Diagnosed data pipeline bottlenecks and optimized **Apache Airflow DAG scheduling and Apache Hive transformations**, improving platform availability by **~40 minutes per day**.
- Led **large-scale client data cleanup and validation efforts**, aligning stakeholders on data requirements and improving data reliability for downstream planning workflows.

Senior Technical Consultant (Data Engineering)

o9Solutions – Bengaluru

Apr 2023 – Mar 2025

- Designed and implemented distributed ETL pipelines using **PySpark, Hive, Python, and Airflow**.
- Processed and validated **100M+ records per week** using distributed transformations.
- Enhanced pipeline efficiency, cutting ETL execution time **by ~30 minutes** per run through targeted query and workflow optimizations.
- Implemented automated monitoring and alerting, ensuring **>99% SLA compliance**.
- Improved platform adoption by **25% by resolving recurring data quality** issues and optimizing workflows.

Technical Consultant (Data Engineering)

o9Solutions – Bengaluru

Jul 2021 – Mar 2023

- Built and maintained batch data pipelines using Hadoop, Hive, Airflow, Python, and SQL.
- Reduced **batch processing time by ~55 minutes** through distributed query optimization and workflow redesign.
- Improved data processing efficiency by **30%** and reduced infrastructure costs by **20%**.
- **Resolved 50+ production pipeline failures** through root cause analysis and long-term reliability improvements.
- Managed end-to-end lifecycle of enterprise data integrations (ingestion → transformation → validation).

Skills

- **Programming:** Python, SQL, Shell
- **Big Data:** Pyspark, Hadoop, Hive
- **Workflow Orchestration:** Apache Airflow

- **ETL Tools:** SSIS
- **Databases:** SQL Server
- **Data Engineering:** ETL Pipelines, Data Modeling, Data Transformation, Data Ingestion
- **DevOps & Tools:** Git, CI/CD, Linux
- **Cloud:** GCP (Exposure)

Education

Bachelor of Technology – Electronics & Communication Engineering

Indian Institute of Information Technology Pune (IIIT Pune)

2017 – 2021

Awards

o9 Spotlight Awards – o9 Solutions

- Recognized by VP of Client Success for resolving a critical production issue and delivering exceptional client support.
- Designed a **centralized ETL workflow framework adopted across 30+ enterprise projects**.
- Received **multiple spotlight awards** for engineering excellence (Jul 2023, Nov 2023, May 2024, Aug 2025).